Assignment -04

Course Details

Course Title: Computer Networks

Couse Code: CSE 405

Section: 05

Submitted To

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➢ Introduction

In this networking, routers play a very important role in connecting different networks and allowing devices from separate locations to communicate with each other. It focuses on designing and configuring a network using three routers connected in a triangular topology. All networks in this project use Class A IP addresses and static routing protocol are applied to manually define the path for data packets. Static routing is a simple and reliable routing method where the network administrator sets the routes manually instead of using any dynamic routing protocols.

Here, three different LANs are connected to three separate routers and the routers themselves are linked using serial DCE connections. After assigning proper IP addresses, configuring interfaces and setting up static routes, communication between PCs located in different networks is tested using the ping command.

➢ Topology

Three routers (Router0, Router2, Router1) connected in a triangular topology:

- Router0 to Router1: serial network 21.0.0.0
- Router0 to Router2: serial network 42.0.0.0
- Router2 to Router1: serial network 63.0.0.0

➢ Three LANs:

- Left LAN (connected to Router0): 84.0.0.0 (PCs: 84.0.0.1, 84.0.0.2)
- Centre LAN (connected to Router2): 105.0.0.0 (PCs: 105.0.0.1, 105.0.0.2)
- Right LAN (connected to Router1): 126.0.0.0 (PCs: 126.0.0.1, 126.0.0.2)

➢ Router & Interface IP Address Table

Device	Interface	IP Address	Subnet Mask
Router0 (Left)	Fa0/0	84.0.0.254	255.0.0.0
	Se3/0	42.0.0.1	255.0.0.0
	Se2/0	21.0.0.1	255.0.0.0
Router2 (Centre)	Fa0/0	105.0.0.254	255.0.0.0
	Se2/0	42.0.0.2	255.0.0.0

	Se3/0	63.0.0.1	255.0.0.0
Router1 (Right)	Fa0/0	126.0.0.254	255.0.0.0
	Se3/0	63.0.0.2	255.0.0.0
	Se2/0	21.0.0.2	255.0.0.0

Table 1: Router Interface Addressing

➢ PC Address Table

LAN	PC	IP Address	Subnet Mask	Default Gateway
Left LAN	PC0	84.0.0.1	255.0.0.0	84.0.0.254
	PC1	84.0.0.2	255.0.0.0	84.0.0.254
Centre LAN	PC2	105.0.0.1	255.0.0.0	105.0.0.254
	PC3	105.0.0.2	255.0.0.0	105.0.0.254
Right LAN	PC4	126.0.0.1	225.0.0.0	126.0.0.254
	PC5	126.0.0.2	255.0.0.0	126.0.0.254

Table 2: End Device IP Configuration

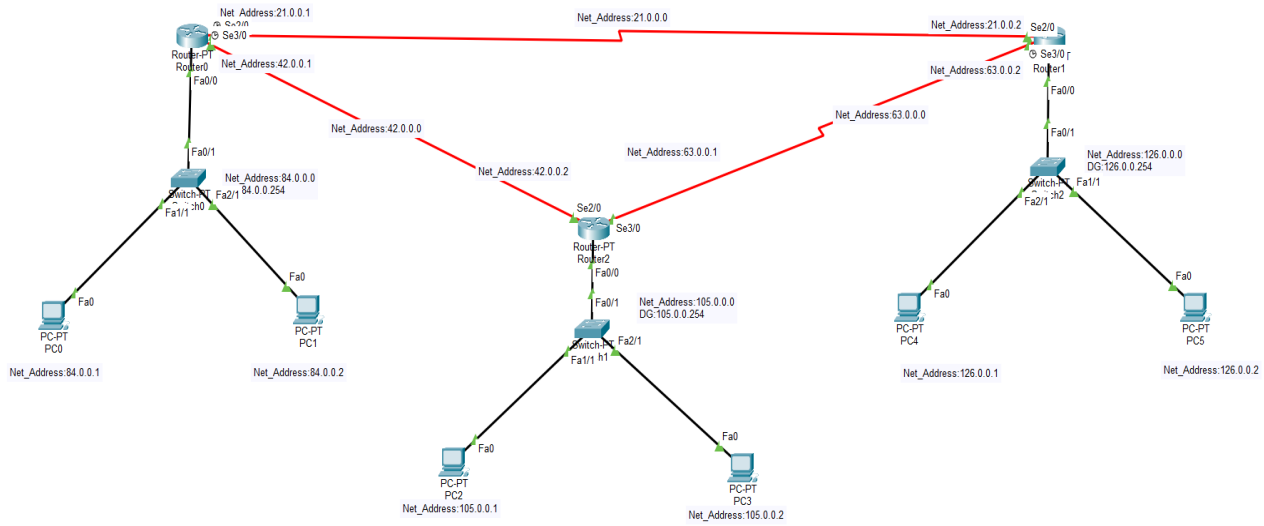

Figure: Network Topology with Three Routers Using Class A IP Addressing and Static Routing

➢ Router Configurations

For Router 0:

se 2/0

Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 21.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 21.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Se 3/0

Router(config)#int se3/0
Router(config-if)#ip address 42.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se3/0
Router(config-if)#ip address 42.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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fa 0/0

Router(config)#int fa0/0
Router(config-if)#ip address 84.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 84.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 21.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se 3/0
Router(config-if)#ip address 42.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 84.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Figure: Router0 configuration

Router0 (Static Configuration)

Router(config)#ip route 105.0.0.0 255.0.0.0 42.0.0.2
Router(config)#ip route 63.0.0.0 255.0.0.0 42.0.0.2
Router(config)#ip route 126.0.0.0 255.0.0.0 21.0.0.2
Router(config)#exit

```
Router(config)#ip route 105.0.0.0 255.0.0.0 42.0.0.2
Router(config)#ip route 63.0.0.0 255.0.0.0 42.0.0.2
Router(config)#ip route 126.0.0.0 255.0.0.0 21.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

For Router 1

se2/0

Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 21.0.0.2 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 21.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#
```

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Se 3/0

Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.2 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.2 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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fa 0/0

Router(config)#int fa0/0
Router(config-if)#ip address 126.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 126.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 21.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.2 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 126.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Figure: Router1 configuration

Router1 (Static Configuration)

Router(config)#ip route 84.0.0.0 255.0.0.0 21.0.0.1
Router(config)#ip route 42.0.0.0 255.0.0.0 21.0.0.1
Router(config)#ip route 105.0.0.0 255.0.0.0 21.0.0.1
Router(config)#exit

```
Router(config)#ip route 84.0.0.0 255.0.0.0 21.0.0.1
Router(config)#ip route 42.0.0.0 255.0.0.0 21.0.0.1
Router(config)#ip route 105.0.0.0 255.0.0.0 21.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

For Router 2

se2/0

Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 42.0.0.2 255.0.0.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 42.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#
```

Se3/0

Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.1 255.0.0.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)#
```

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fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 105.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 105.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Copy Paste

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 42.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#int se 3/0
Router(config-if)#ip address 63.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)#int fa0/0
Router(config-if)#ip address 105.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

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Figure: Router2 configuration

Router2 (Static Configuration)

Router(config)#ip route 21.0.0.0 255.0.0.0 42.0.0.1

Router(config)#ip route 84.0.0.0 255.0.0.0 42.0.0.1

Router(config)#ip route 126.0.0.0 255.0.0.0 63.0.0.2

Router(config)#exit

```
Router(config)#ip route 21.0.0.0 255.0.0.0 42.0.0.1
Router(config)#ip route 84.0.0.0 255.0.0.0 42.0.0.1
Router(config)#ip route 126.0.0.0 255.0.0.0 63.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

➢ Device Configuration

For PC 0

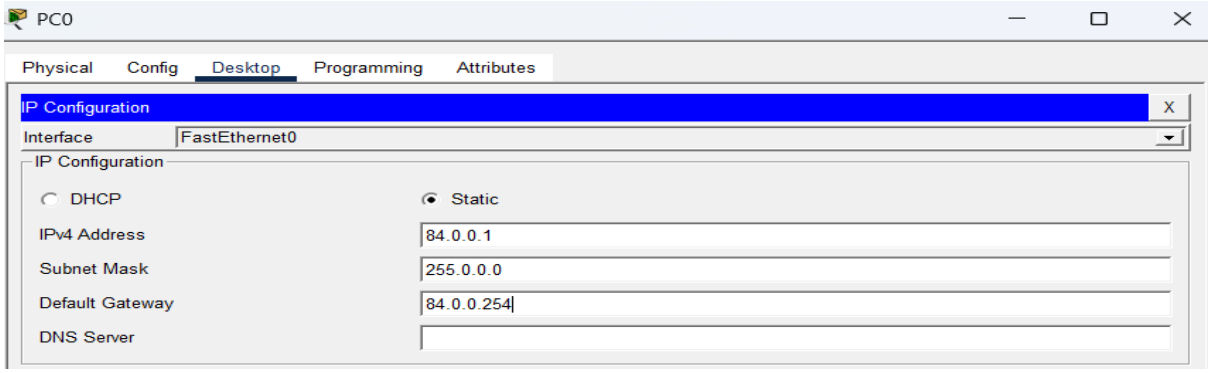

Figure: PC Configuration of PC0

For PC 1

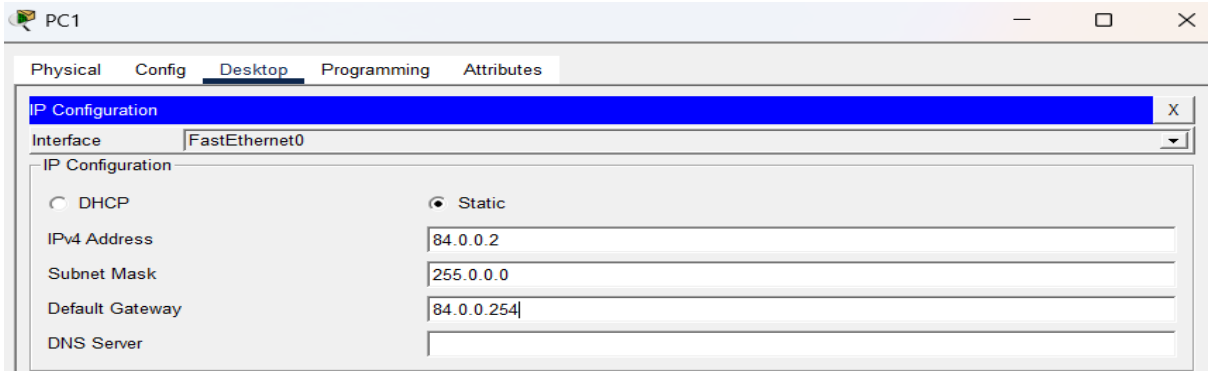

Figure: PC Configuration of PC1

For PC 2

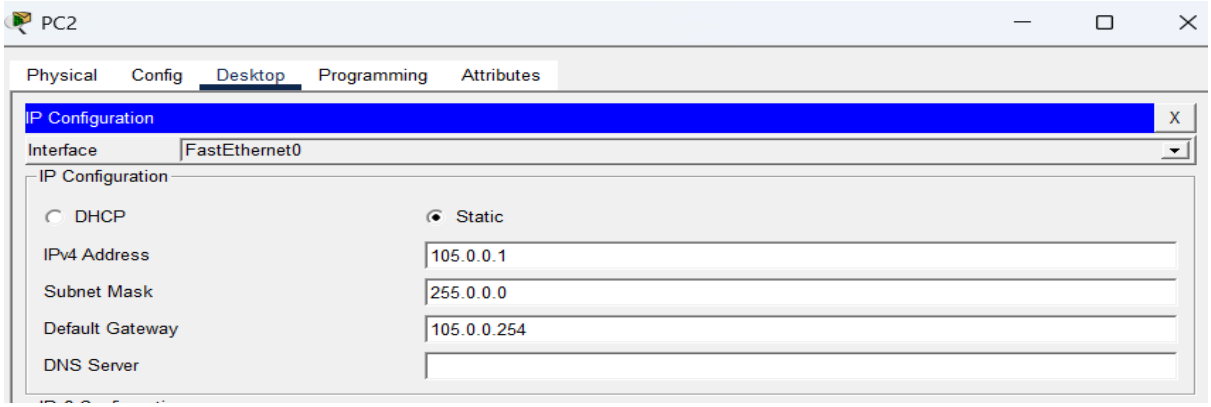

Figure: PC Configuration of PC2

For PC 3

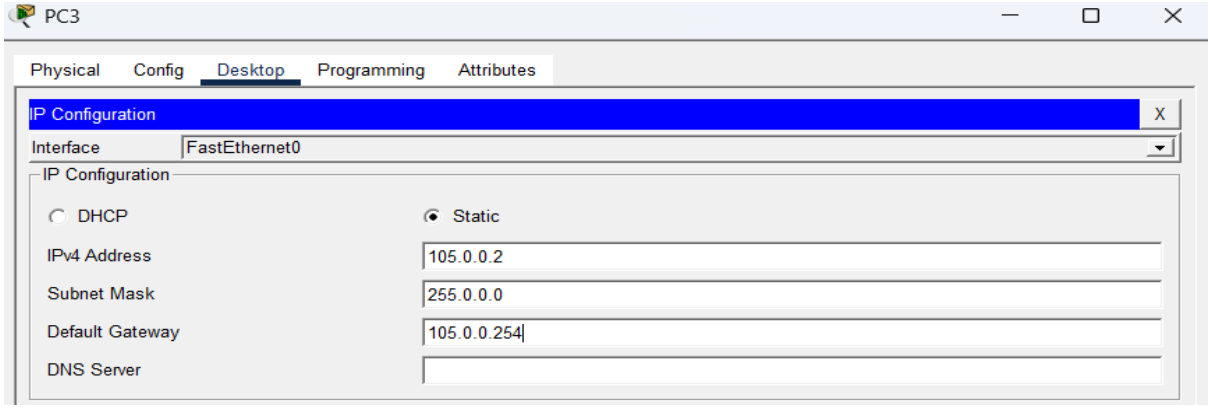

Figure: PC Configuration of PC3

For PC 4

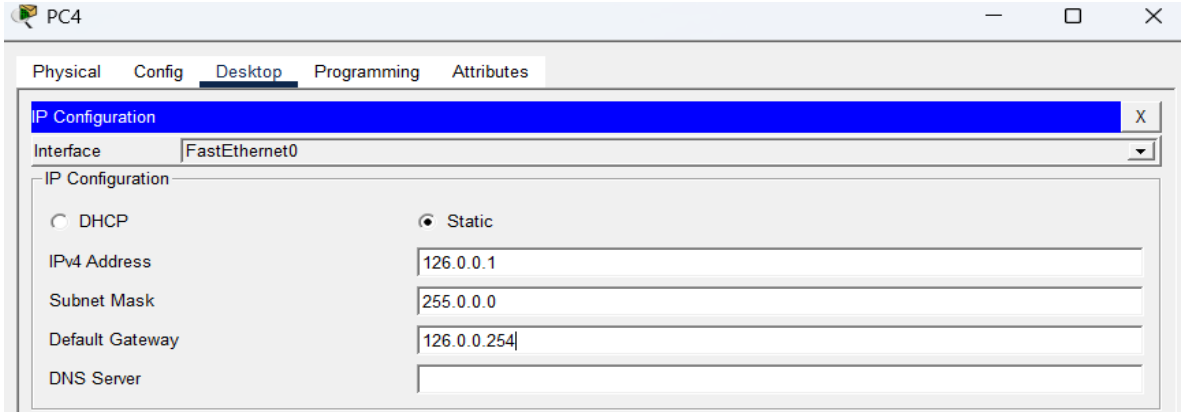

Figure: PC Configuration of PC4

For PC5

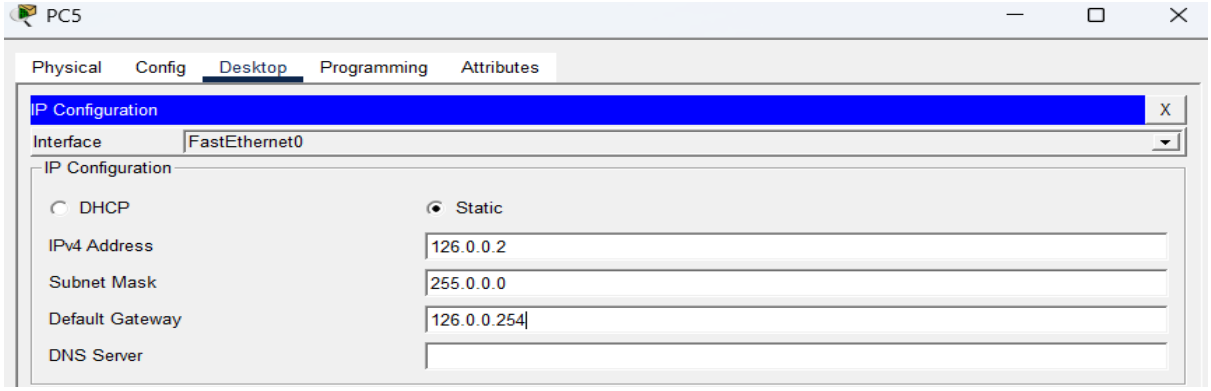

Figure: PC Configuration of PC5

➢ Ping Communication

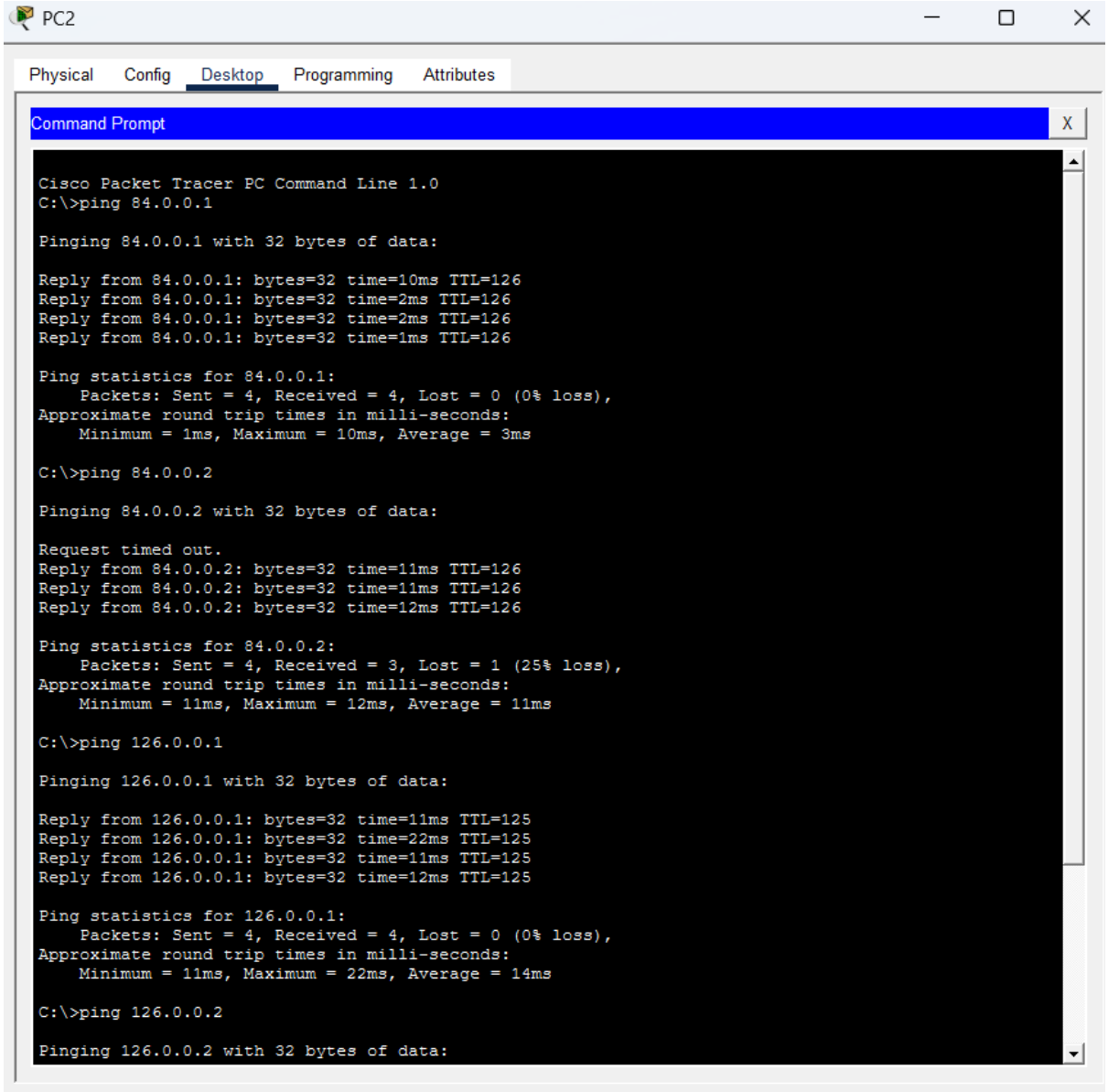

Figure: Ping Command

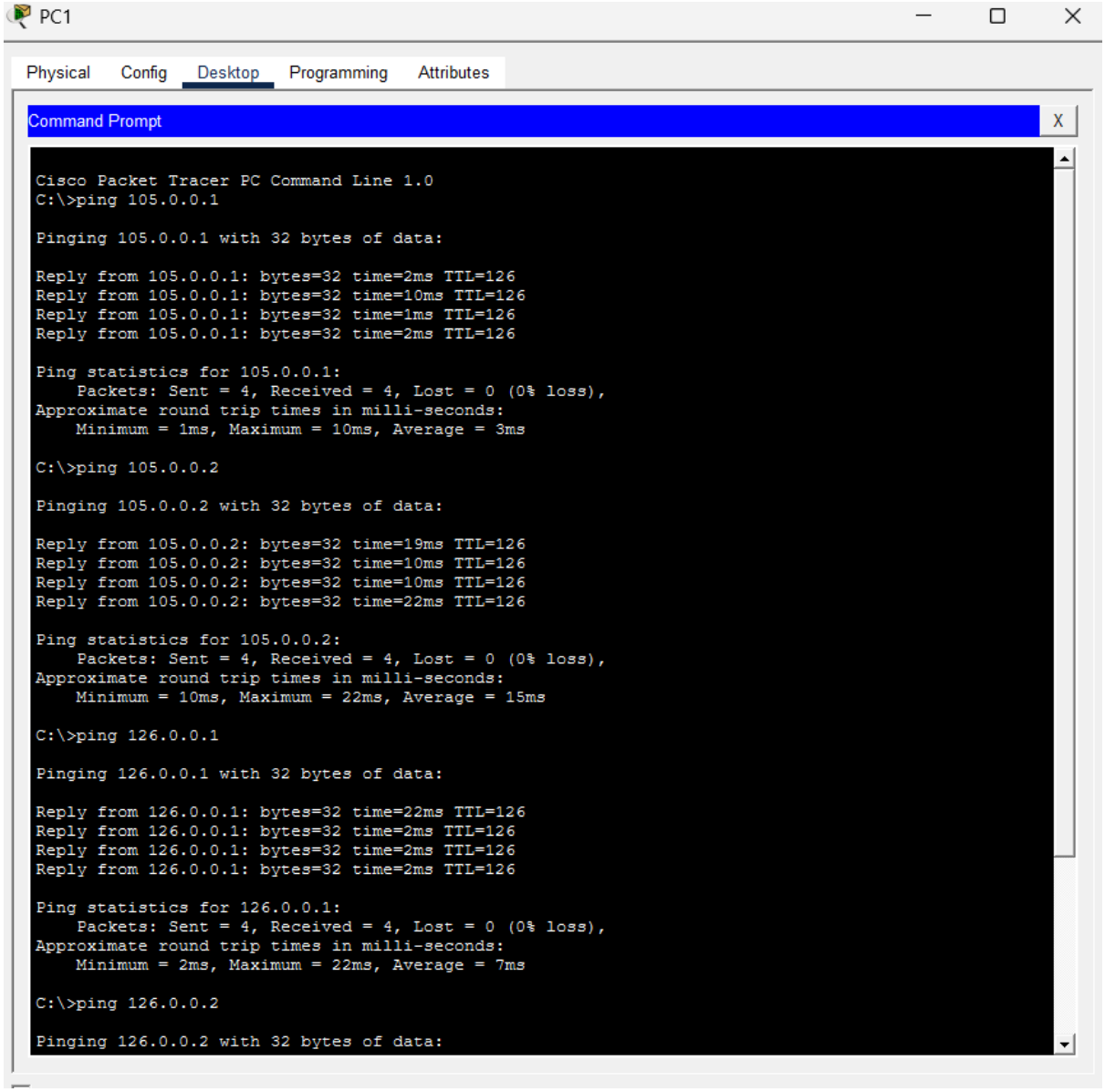

Figure: Ping Command

➢ Test Results Between PCs

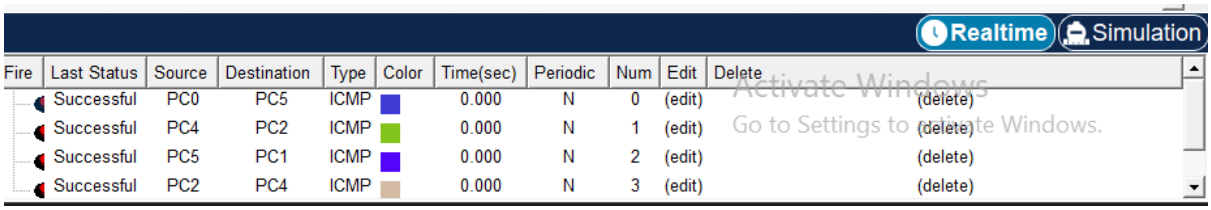

Figure: Ping Test Results Between PCs

➢ Conclusion

In this network, a three router network topology was successfully designed and configured using Class A IP addressing for all networks. Each router was assigned proper interface IPs, subnet masks, clock rates (for serial links and static routes to ensure full end to end connectivity. All PCs in the left, centre and right LANs were configured with valid IP addresses, subnet masks and default gateways according to the network design.

After completing the static routing configuration on all three routers, connectivity testing was performed using the ping command from PCs located in different networks. The test results showed successful ping replies proving that the static routes were correctly implemented and that all routers could properly forward packets across the network.This confirms that the entire network is fully functional, all routing paths are properly established, and communication between distant LANs is working without any errors.